



PROJECT KNOT

USER INFORMATION MANUAL

*Centralized Academic Resource Reservation & Campus Maintenance Management System
Modeled under University Corporate Operational Standards*

Document Identification:	ENG/PDN/KNOT/UIM/2026-V2.5 (Gold Master Release)
Standard Operating Audience:	Students, Academic Lecturers, Booking Admins (AR Office), Works Engineers, Technicians
Governing Institution:	Faculty of Engineering, University of Peradeniya, Sri Lanka
Live Web Deployment:	Production: https://foe.knotpdn.tech
Security Classification:	Authorized Academic & Maintenance Personnel (Single Sign-On SSO Protected)

SECTION OVERVIEW: FUNCTION OVERVIEW & CREDENTIALS MATRIX

General Description of Functionality

Project KNOT (University Resource & Maintenance Management Platform) is an integrated enterprise-grade portal engineered specifically for the Faculty of Engineering at the University of Peradeniya. The system replaces fragmented paper slips and disparate messaging channels by uniting two vital institutional workflows: (1) Academic Resource Reservation for the standardized 10 faculty venues with an hourly 08:00–18:00 availability grid, strict lecturer academic endorsement, and an automated 2-step verification anti-no-show engine; and (2) Infrastructure Maintenance Reporting with Leaflet/OpenStreetMap GIS pin drops, 50MB base64 photographic evidence capture, priority classification dispatch, and proof-of-work completion logging.

System Demonstration Access Credentials Matrix

For quality assurance verification, faculty committee evaluation, and administrative testing, pre-configured role accounts are provisioned below with pre-seeded database records:

Role Category	User ID	Access Key	Assigned Identity	Core Operating Scope
Student Booker	e22237	1234	Minhaj Ali	Hall reservation grid, 2-step verification confirmation, GIS fault reporting
Academic Lecturer	lecturer1	1234	Dr. Smith	Direct hall booking, student request endorsement/rejection, faculty timetable
Booking Admin (AR)	bookadmin	adminpass	AR Office Staff	Final reservation approval queue, multi-filter search, CSV master importer
Maintenance Admin	admin	adminpass	Works Engineer	Central ticketing dashboard, evidence inspector, priority dispatch to workers
Field Technician	alex	1234	Alex Johnson	Assigned task queue, interactive directive viewer, proof-of-work photo log

Manual Organization & Table of Contents

This user manual is organized into modular role-based sections following corporate banking documentation standards. Each section provides functional context, step-by-step procedures, and high-resolution interface figures:

Section	Functional Operational Scope	Primary UI Figures	Reference Page
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SECTION 1: SYSTEM OVERVIEW & SECURE PORTAL ACCESS

General Description of Functionality

The Project KNOT Single Sign-On (SSO) Portal provides a centralized, secure authentication boundary protecting academic and facility resources. Any valid student registration number, academic staff username, or administrative service account dynamically routes the authenticated user to their role-specific dashboard. The system incorporates secure session tokens, role-based access control (RBAC), and automated route protection.

General Description of Solution & Step-by-Step Procedure

Step 1 – Open Web Browser: Launch Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari on desktop or tablet.

Step 2 – Navigate to URL: Enter <https://foe.knotpdn.tech> (Production) or <http://localhost:3000> (Local Environment).

Step 3 – Input Account Credentials: Enter your assigned University User ID (e.g. e22237, lecturer1, bookadmin, admin, or alex) and Password.

Step 4 – Select Role & Sign In: Click Sign In. The platform automatically resolves your authorized security privileges and launches your designated portal interface.

CRITICAL REQUIREMENT: Session Security & Active Token Timeout

Never share credentials. To maintain faculty data integrity, sessions automatically terminate after periods of inactivity. Always click the Logout button located under your avatar in the upper right header when leaving public computer workstations.

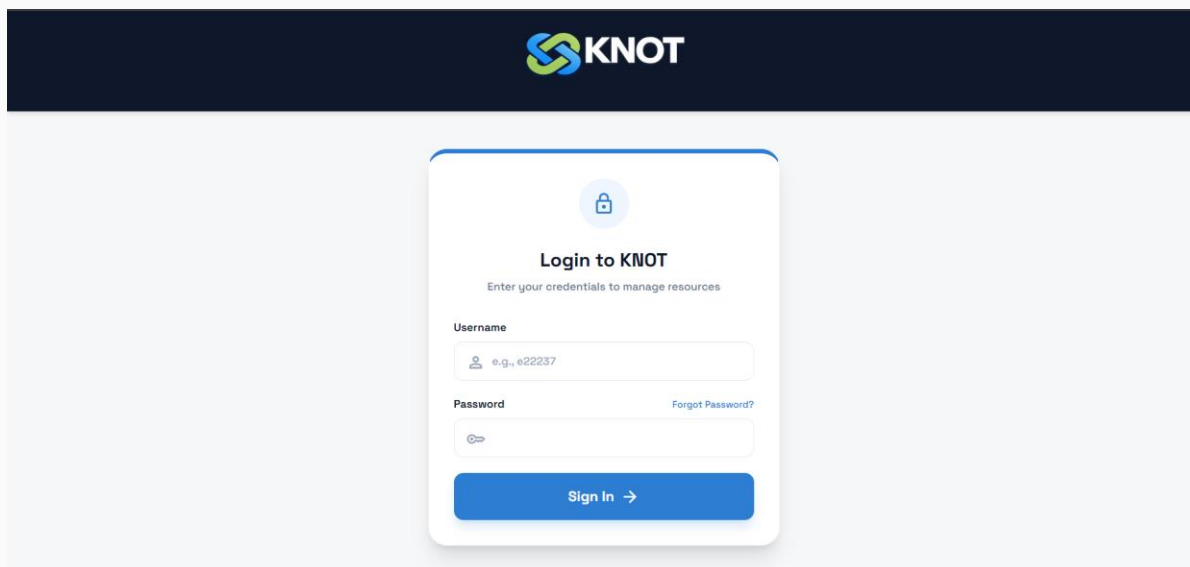


Figure 1.1: Project KNOT Unified Single Sign-On (SSO) Portal

SECTION 2: USER ACCOUNT & PROFILE MANAGEMENT

General Description of Functionality

Each registered member of the faculty possesses a dedicated user profile reflecting their official academic affiliation, departmental assignment, contact phone number, university email address, and profile avatar. Keeping contact details current is essential because automated notifications—such as 2-step verification prompts and maintenance job dispatches—are routed directly to the designated email address.

General Description of Solution & Step-by-Step Procedure

Step 1 – Access User Menu: Click on your profile avatar icon situated in the top right-hand corner of the main navigation bar.

Step 2 – Select Profile Option: Click 'Profile' from the dropdown menu to open the interactive Profile Details modal dialog.

Step 3 – Update Contact Details: Modify your Email Address, Contact Phone Number, Registration/Employee Number, Department, or Campus Address.

Step 4 – Save Profile Changes: Click 'Save Changes'. The updated information is immediately synchronized across the central database.

INFORMATION: Profile Avatar & Departmental Verification

Staff and students can upload photo avatars up to 5MB. Departmental assignments (e.g. Civil, Mechanical, Electrical, Computer) determine the timetable calendar filtering scopes available on the Lecturer Dashboard.

User Interface Design & Description: Profile Modals across Roles

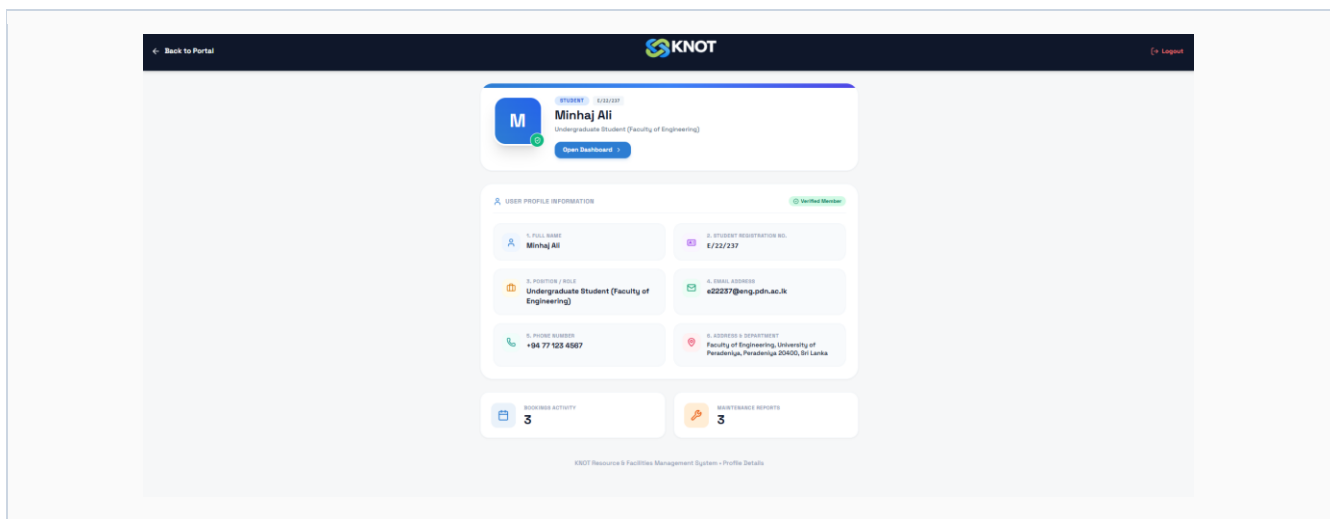


Figure 2.1: Student User Profile & Academic Contact Details Modal

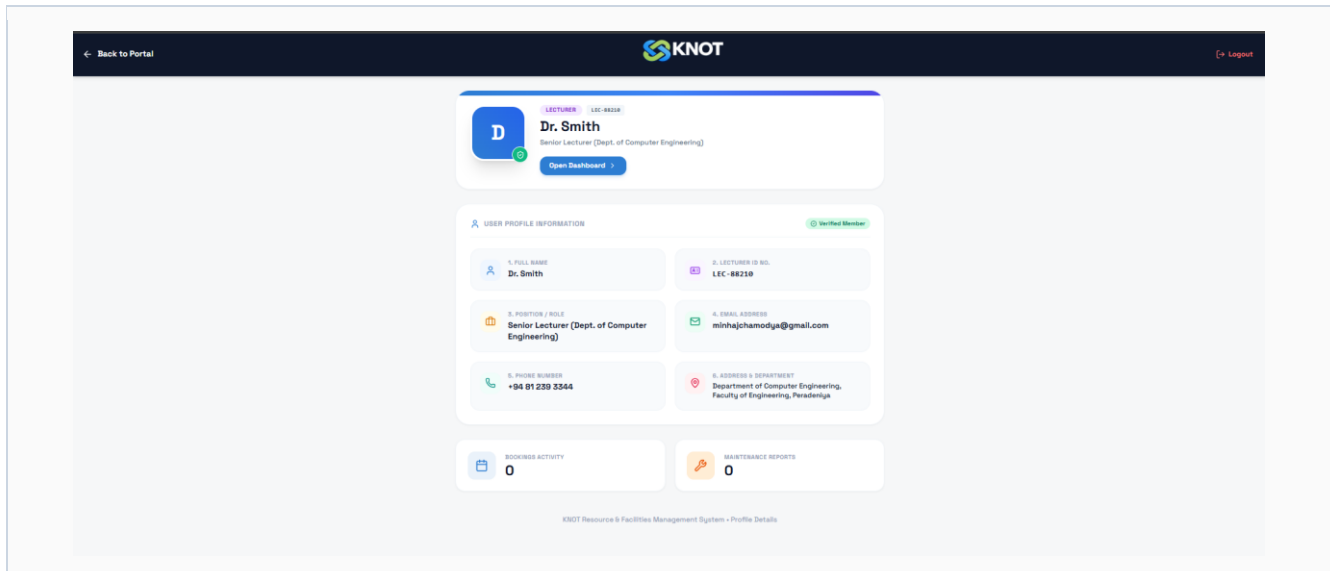


Figure 2.2: Lecturer Academic Staff Profile Modal

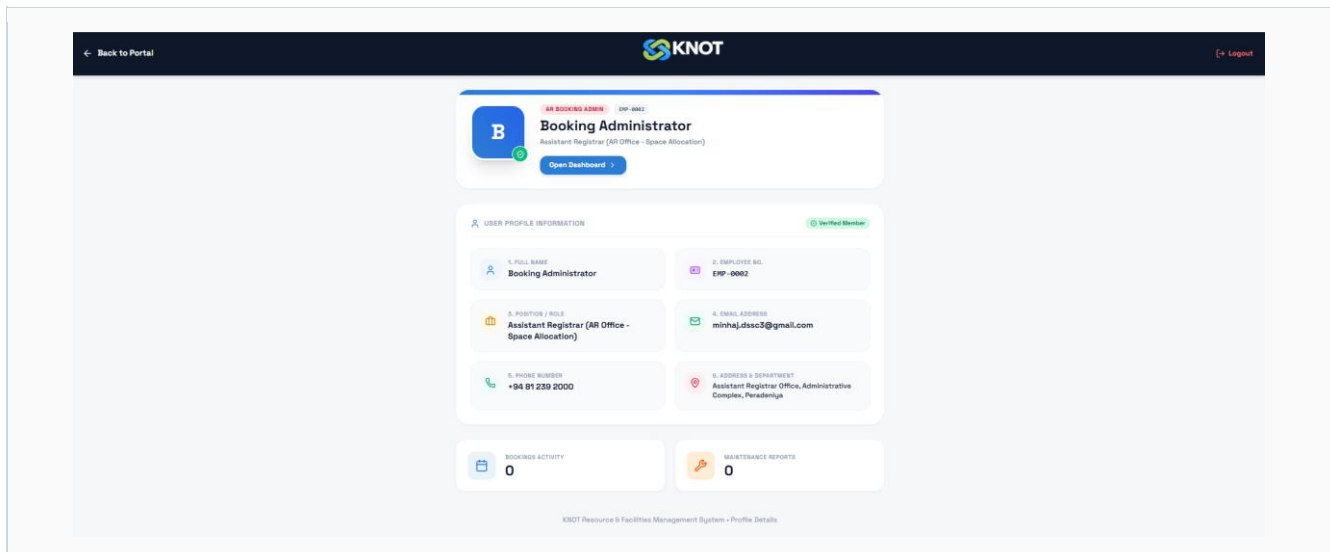
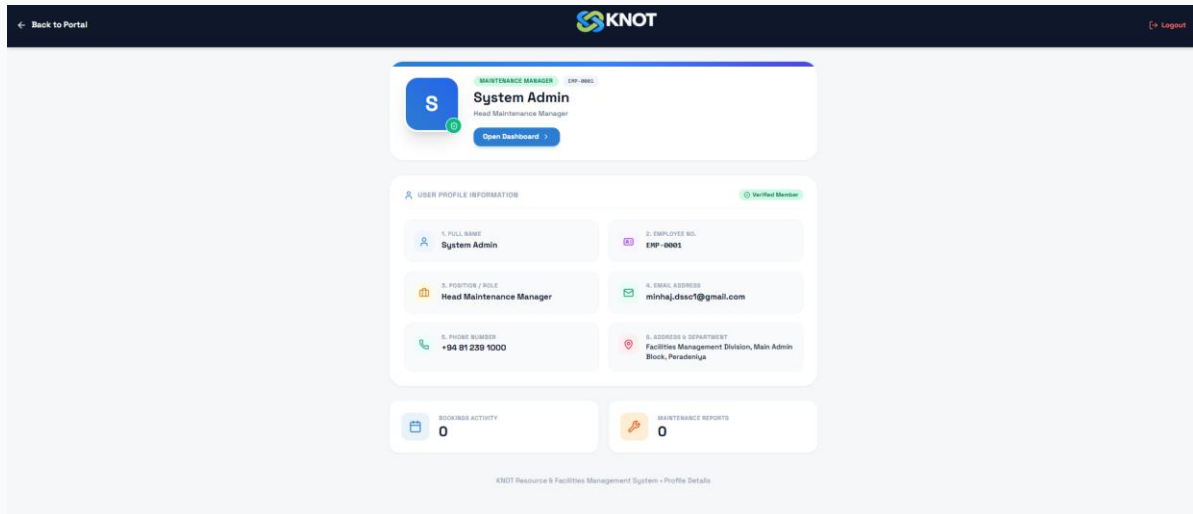
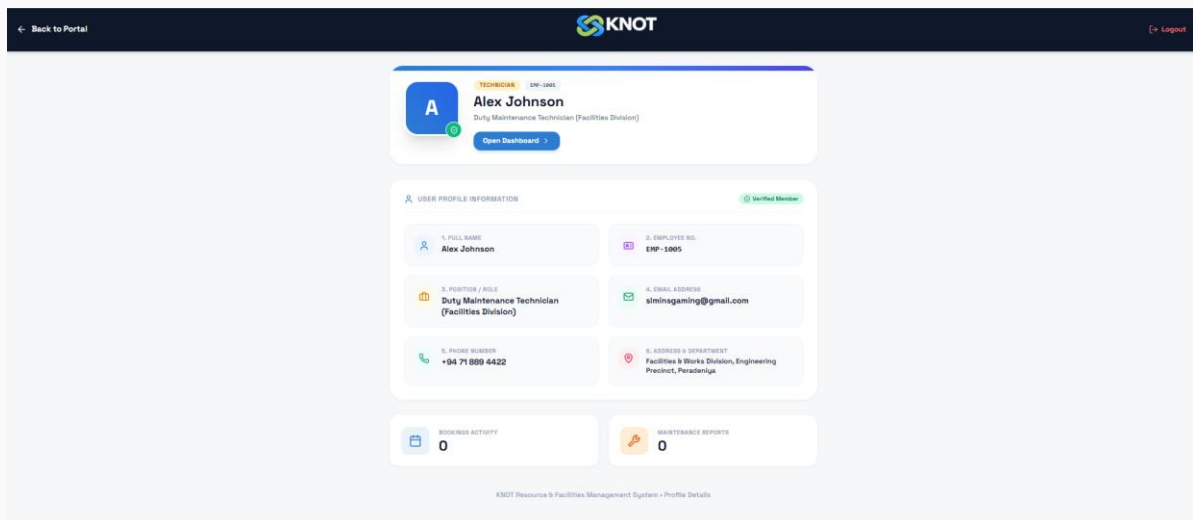


Figure 2.3: Assistant Registrar (AR Booking Admin) Profile Modal

**Figure 2.4: Maintenance Works Engineer Admin Profile Modal****Figure 2.5: Maintenance Field Technician Profile Modal**

SECTION 3: STUDENT PORTAL USER GUIDE

General Description of Functionality

The Student Portal enables undergraduate and postgraduate engineering students to perform self-service academic reservations across 10 official halls and labs, manage reservation statuses in real-time, comply with the mandatory 2-step attendance verification anti-no-show protocol, and submit geotagged maintenance fault reports.

3.1 Navigating the Student Dashboard

General Description of Functionality

Upon authentication, the Student Dashboard (e22237) presents a cohesive view divided into key operational widgets: Active Bookings summary cards with live status badges (Approved, Pending Lecturer, Pending AR, Rejected), the 2-Step Verification Action Banner prompting for immediate attendance confirmation, and My Reported Faults tracking the lifecycle of submitted infrastructure tickets.

General Description of Solution & Step-by-Step Procedure

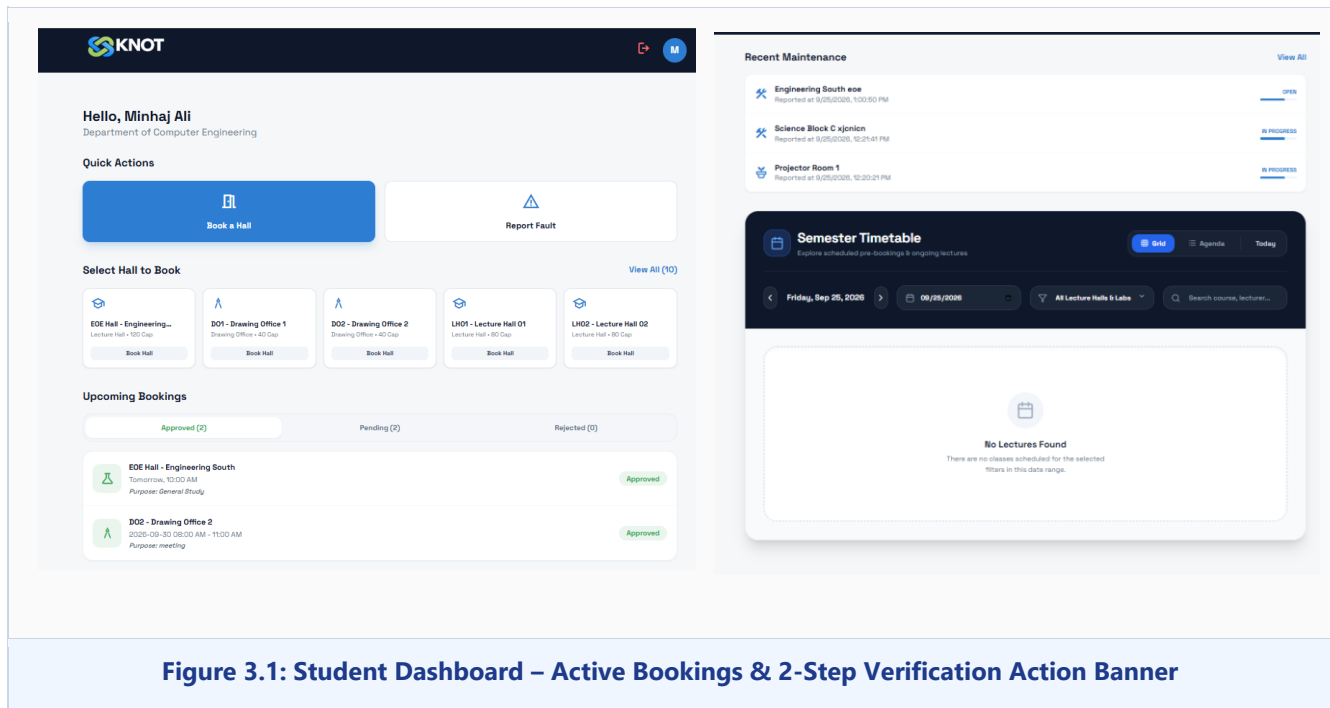
Step 1 – Review Active Bookings: Inspect existing reservations, assigned venue names, dates, hourly intervals, and endorsing lecturers.

Step 2 – Monitor Status Pills: Green badges indicate official AR approval; yellow pills denote pending endorsements; red badges indicate rejection.

Step 3 – Track Maintenance Reports: Review progress logs on campus faults you have submitted, including technician work updates.

INFORMATION: Multi-Part Interface Presentation

Due to the extensive informational density of the dashboard, the interface is displayed below in two sequential sections: Part 1 covers active bookings and immediate action banners; Part 2 covers reported fault logs and quick-action navigation controls.



3.2 Hall & Lab Reservation Workflow (Hourly Availability Grid)

General Description of Functionality

KNOT standardizes all reservations into 1-hour discrete slots between 08:00 AM and 06:00 PM. Students can browse availability across 10 standardized faculty venues: EOE Hall, DO1 (Drawing Office 1), DO2, LH01, LH02, Seminar Room A, Seminar Room B, Computer Lab 01, Computer Lab 02, and Electronics Lab.

General Description of Solution & Step-by-Step Procedure

Step 1 – Initiate Booking: Click 'Book Space' from the top navigation bar or the dashboard quick action button.

Step 2 – Select Reservation Date: Choose the target booking date using the calendar date picker control.

Step 3 – Select Venue / Room: Choose the desired venue from the standardized 10-room dropdown menu.

Step 4 – Select Hourly Slots: Interact with the Hourly Availability Grid: click one or multiple contiguous Green badges (Available). Red badges represent occupied slots.

Step 5 – Assign Supervising Lecturer: Select your supervising academic lecturer from the dropdown menu (e.g. Dr. Smith). Student bookings require academic endorsement.

Step 6 – Specify Academic Purpose: Enter concise course or project details (e.g. 'CO2060 Project Demonstration and Hardware Integration Testing').

Step 7 – Submit Request: Click 'Submit Booking Request'. The reservation status enters 'Pending Lecturer Endorsement'.

Figure 3.3: Interactive 10-Venue Hourly Reservation Grid & Purpose Submission

3.3 2-Step Attendance Verification System (Anti-No-Show Engine)

General Description of Functionality

To maximize utilization of scarce academic spaces and prevent lecture halls from remaining vacant due to unreleased bookings, Project KNOT enforces an automated 2-Step Attendance Verification Protocol powered by a background chronometer engine.

General Description of Solution & Step-by-Step Procedure

Step 1 – Advance Bookings (>12 Hours Prior): The system sends an email notification and displays an action banner on your dashboard exactly 24 hours prior to the slot start. You must confirm attendance at least 12 hours before start.

Step 2 – Same-Day Bookings (<12 Hours Prior): The action banner appears at 07:00 AM on the morning of the booking. You must click confirmation before the 07:30 AM hard cutoff.

Step 3 – Confirming Attendance: Click the green '✓ Confirm Attendance' button. Your reservation status transitions to 'Verified' and guarantees your venue access.

Step 4 – Releasing Unneeded Slots: If plans change, click 'X Cancel & Release Slot'. The slot is immediately returned to the public pool as Available for fellow students.

⚠ CRITICAL REQUIREMENT: Automatic Cancellation Policy

If neither action is taken before the countdown expiration deadline, the background engine automatically cancels the booking, logs an unverified reservation strike, and marks the hall as available for rebooking.

3.4 Campus Maintenance Fault Reporting (GIS Map Pins & Photo Evidence)

General Description of Functionality

Students act as vigilant observers across campus. The maintenance module empowers any student to report defects (e.g. failed projectors, leaking pipes, non-functional air conditioners, broken desks) directly to the Works Engineering Office.

General Description of Solution & Step-by-Step Procedure

Step 1 – Open Issue Form: Click 'Report Maintenance Issue' from the navigation bar or dashboard button.

Step 2 – Select Fault Category: Choose the appropriate category: Projector/Audio, Air Conditioning, Electrical, Plumbing, or Structural/Furniture.

Step 3 – Provide Concise Title & Description: Enter a specific title (e.g. 'EOE Hall East Projector HDMI Signal Failure') and contextual description.

Step 4 – Drop GIS Pin on OpenStreetMap: Click on the interactive OpenStreetMap Leaflet interface. The system captures exact GPS coordinates and executes reverse geocoding to determine address text.

Step 5 – Attach Photographic Evidence: Upload a high-resolution photograph of the defect (up to 50MB Base64 format). Visual evidence accelerates engineering triage.

Step 6 – Submit & Track: Click 'Submit Fault Report'. A unique Ticket ID is generated, and automated dispatch notifications are sent.

Figure 3.4: Maintenance Fault Reporting – Category Selection & OpenStreetMap GIS Pin Drop (Part 1)

SECTION 4: LECTURER PORTAL USER GUIDE

General Description of Functionality

The Lecturer Portal provides academic staff (lecturer1) with dual operational capabilities: (1) Serving as the academic authority to review and endorse or reject student hall reservations, and (2) Direct reservation privileges for lectures, guest seminars, and mid-semester examinations bypassing student triage, alongside full department calendar visibility.

4.1 Reviewing & Endorsing Student Booking Requests

General Description of Solution & Step-by-Step Procedure

Step 1 – Access Endorsement Queue: Log in as an academic lecturer. The Pending Approvals widget lists all student reservation requests naming you as supervisor.

Step 2 – Evaluate Request Merits: Inspect student registration number, designated venue, purpose narrative, and requested hourly slot intervals.

Step 3 – Grant Endorsement: Click '✓ Endorse Request'. The reservation status immediately advances to 'Pending AR Approval' in the Assistant Registrar's queue.

Step 4 – Reject Request: Click 'X Reject Request' and enter constructive feedback notes (e.g. 'Hall DO1 reserved for postgraduate examinations; please re-book DO2').

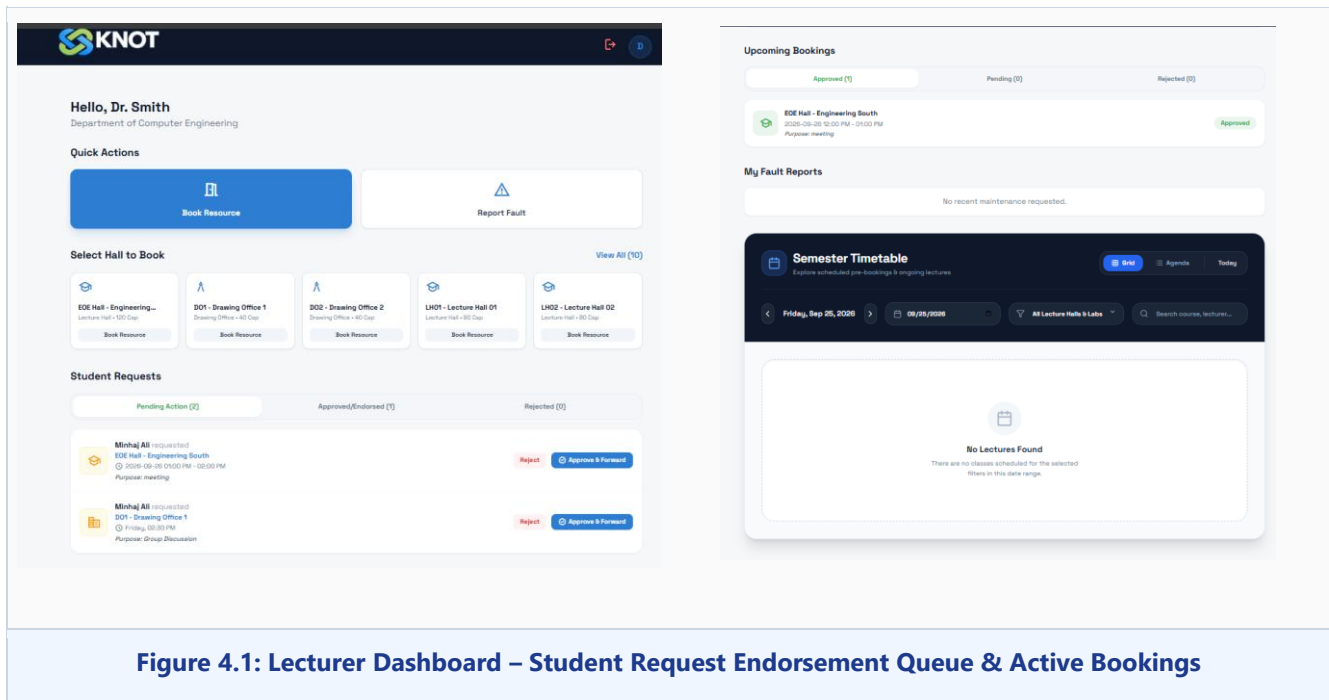


Figure 4.1: Lecturer Dashboard – Student Request Endorsement Queue & Active Bookings

4.2 Department Schedule Timeline & Hall Calendar Inspection

General Description of Functionality

Lecturers can monitor the chronological timeline of lecture halls and drawing offices to prevent scheduling conflicts when planning supplementary tutorial classes or student project evaluations.

General Description of Solution & Step-by-Step Procedure

Step 1 – Filter by Date Range: Select target academic dates or switch between Day, Week, and Month timetable calendar views.

Step 2 – Filter by Venue or Department: Isolate specific facilities (e.g. EOE Hall or Computer Lab 01) to verify upcoming course sessions.

Step 3 – Direct Room Booking: Click 'Book Direct Space'. Lecturer direct reservations bypass supervisory endorsement and enter the AR Office queue directly (or receive instant confirmation if Auto-Booking is active).

SECTION 5: BOOKING ADMINISTRATOR (AR OFFICE) GUIDE

General Description of Functionality

The Booking Administrator (bookadmin) represents the Assistant Registrar's (AR) Office, holding ultimate institutional authority over the allocation and policy enforcement of all faculty infrastructure. Key administrative responsibilities include reviewing lecturer-endorsed requests, toggling the automated booking approval engine, conducting multi-criteria audit searches, and importing the master semester timetable CSV.

5.1 AR Approval Queue & Decision Processing

General Description of Solution & Step-by-Step Procedure

Step 1 – Open AR Management Queue: Log in as 'bookadmin'. The central approval queue displays all requests requiring institutional authorization.

Step 2 – Review Comprehensive Dossier: Review student credentials, endorsing lecturer notes, purpose statement, and requested time boundaries.

Step 3 – Grant Final Approval: Click 'Approve'. The booking enters 'Approved' status, an automated receipt is dispatched via email, and the slot is locked on the public calendar.

Step 4 – Issue Rejection: Click 'Reject' and supply mandatory administrative reason codes (e.g. 'Annual Faculty Board Meeting scheduled').

5.2 Advanced Multi-Criteria Filter & Search Engine

General Description of Functionality

The AR Dashboard incorporates a high-performance filtering toolbar allowing administrators to isolate records across thousands of entries: by Student Reg No, Lecturer Name, Room Badge (EOE, DO1, DO2, LH01, LH02, etc.), Date Presets (Today, Tomorrow, This Week, Future), and Booking Status (Approved, Pending AR, Pending Lecturer, Rejected, Cancelled).

5.3 Master Semester Timetable Bulk CSV Importer & Auto-Booking Toggle

General Description of Solution & Step-by-Step Procedure

Step 1 – Configure Auto-Booking Engine: Toggle the 'Automated Booking Engine' switch in the upper toolbar. When set to 'ON', incoming conflict-free requests receive immediate approval without manual queuing.

Step 2 – Prepare Semester CSV File: Ensure columns conform to standard schema: Room, Day, StartTime, EndTime, CourseCode, Lecturer.

Step 3 – Upload & Execute Import: Click 'Bulk Schedule Import', select the master schedule file, and click 'Import Timetable'. The validation engine checks for venue collisions and populates recurring weekly lectures across the semester.

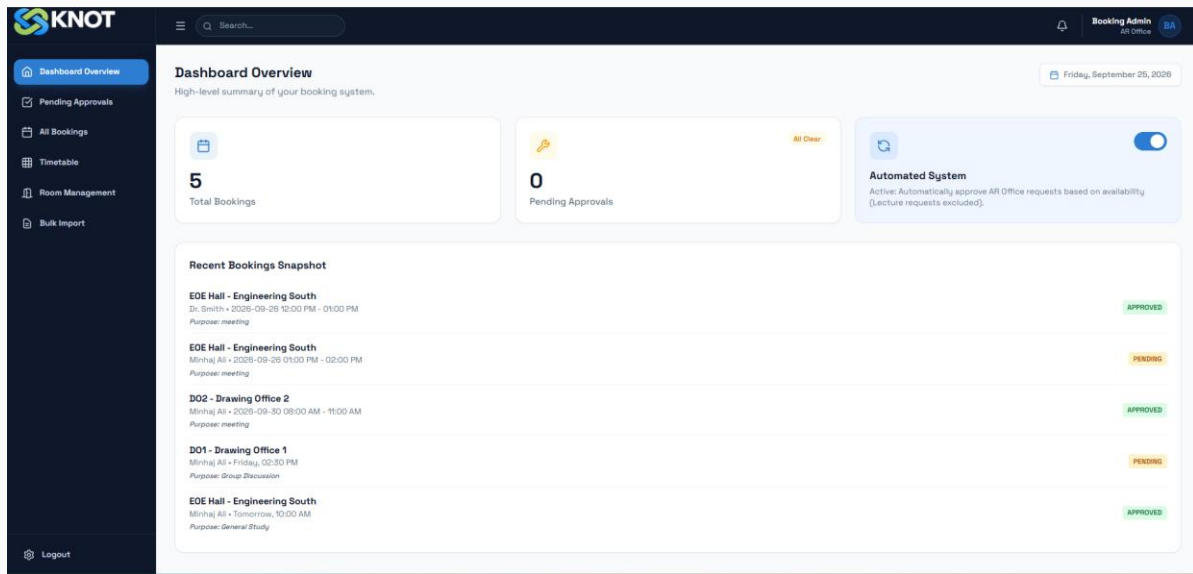


Figure 5.1: AR Office Master Dashboard – Approval Queue, Advanced Multi-Filtering & Master Timetable Importer

SECTION 6: MAINTENANCE ADMINISTRATOR GUIDE

General Description of Functionality

The Maintenance Administrator (admin) represents the Faculty Works Engineering Department. The administrator oversees campus physical plant integrity, assesses reported fault tickets, inspects visual photographic evidence, classifies urgency priorities, and dispatches specialized field technicians.

6.1 Central Ticketing Management Dashboard

General Description of Solution & Step-by-Step Procedure

Step 1 – Monitor Live Ticket Feed: Log in as Maintenance Admin. The dashboard categorizes tickets into Open, In Progress, and Resolved status pillars.

Step 2 – Filter by Trade Category: Filter tickets by electrical, plumbing, HVAC, audiovisual, or carpentry disciplines.

Step 3 – Select Ticket for Triage: Click on any ticket entry to launch the comprehensive Ticket Details and Dispatch modal.

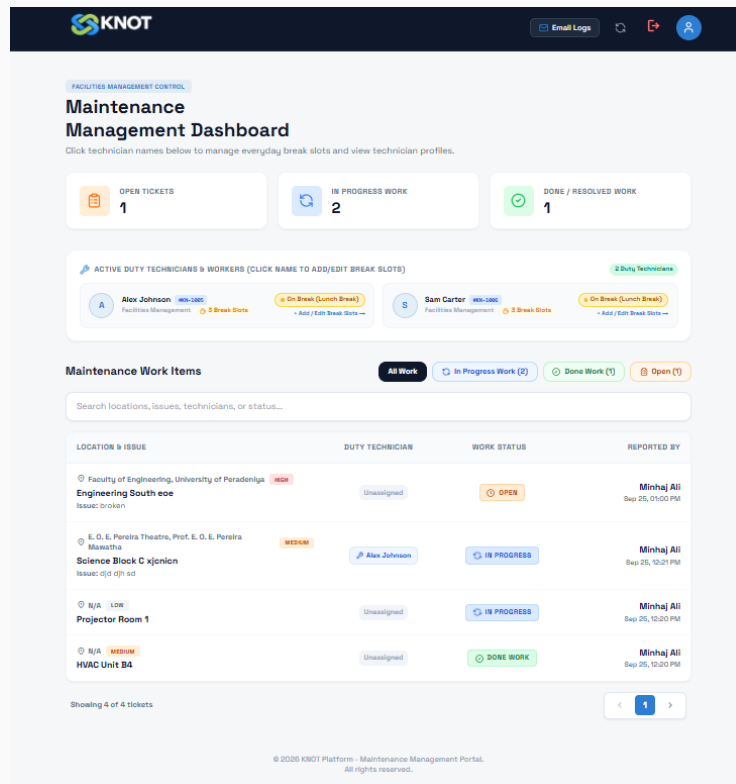


Figure 6.1: Central Maintenance Ticketing Dashboard & Trade Category Status Filters

6.2 Ticket Details Inspection & Technician Task Assignment

General Description of Solution & Step-by-Step Procedure

Step 1 – Inspect Photographic Evidence: View the full-resolution evidence photo uploaded by the reporting student or staff member.

Step 2 – Verify GIS Coordinates: Review the precise latitude/longitude coordinates and campus location description pinpointed on OpenStreetMap.

Step 3 – Assign Field Technician: Select an authorized technician from the staff roster (e.g. Alex Johnson or Sam Carter).

Step 4 – Set Job Priority Level: Classify ticket urgency: Low (routine aesthetic repairs), Medium (scheduled maintenance), High (disrupted lecture), or Urgent (safety/flood/electrical hazard).

Step 5 – Input Managerial Directives: Enter specific execution instructions (e.g. 'Replace 5000-lumen lamp unit; test HDMI signal splitter before signing off').

Step 6 – Dispatch Work Order: Click 'Assign & Update Ticket'. The ticket transitions to 'In Progress' and triggers an automated dispatch notification to the technician's terminal.

Initial Work Circle: Assign Duty Technician
This maintenance request is in the initial stage. Please assign an available duty technician below to dispatch this task. Once the technician completes the work and submits their solution proof, the Manager Review & Verification options will be unlocked.

BTXT-0004 | REPORTED ON SEP 25, 2025, 01:00 PM

Engineering South eoe

LOCATION
Faculty of Engineering, University of Peradeniya, Prof. E. D. E. Perera Mawatha, Peradeniya, Eriyagama, Kandy District, Central Province, 20400, Sri Lanka
Map Coordinates: 7.25437, 80.59222

REPORTED BY
Minhaj Ali

REPORTED ISSUE DETAILS & PHOTO
(0/10) INITIAL PHOTO (REPORTED ISSUE)

REPORTED DESCRIPTION
broken

ASSIGN DUTY TECHNICIAN
Select a qualified technician from the directory below to assign and dispatch this maintenance work order.

Alex Johnson (Facilities Management) [ON BREAK: Lunch Break]

Assigned Technician: Alex Johnson

Manager Verification Controls Locked
Options to Confirm & Verify Resolved or Send Next Resolving Step will activate here automatically once the assigned technician completes work and submits their solution proof.

WORK ORDER PROGRESS

OPEN IN PROGRESS RESOLVED

UPDATE STATUS

OPEN IN PROGRESS RESOLVED

Save & Assign Technician

Figure 6.2: Maintenance Ticket Details & Evidence Inspector

6.3 Technician Solvation Review & Manager Verification Workflow

General Description of Solution & Step-by-Step Procedure

Step 1 – Monitor Solvation Review Queue & In-App Banner: When the assigned technician submits their on-site work log and proof of repair, the system automatically transitions the ticket to 'Pending Manager Verification' and displays a prominent green alert banner: 'Technician Solvation Submitted for Manager Review'. The Maintenance Administrator opens the ticket to initiate formal quality assurance verification.

Step 2 – Inspect Fault Metadata & Reverse-Geocoded Location: Verify core ticket parameters including the unique Ticket ID (e.g. #TKT-0007), submission timestamp, faculty venue (e.g. Main Library), reporting student/staff identity (e.g. Minhaj Ali), OpenStreetMap pinpoint with GPS coordinates (7.25316, 80.59300), and the reporter's initial defect description.

Step 3 – Perform Side-by-Side 'Before & After' Solvation Comparison: Scrutinize the dual-panel resolution comparison inspector:

- Before Solved (Initial Request): Review the original defect photo and initial problem description ('Broke') provided at the time of ticket logging.
- After Solved (Technician Solvation): Scrutinize the proof-of-work photo uploaded by the technician along with their technical resolution log ('Ok' / repairs performed).

Step 4 – Document Manager Assessment & Follow-Up Directives: Enter supervisory comments, QA evaluation notes, or follow-up remediation instructions in the 'Manager Review & Verification Options' text field.

Step 5 – Execute Review Action (Confirm Verification vs. Re-Dispatch): Select the appropriate operational action based on visual evidence evaluation:

- Confirm & Verify Resolved (Green Button): Officially verifies satisfactory completion, updates ticket state to 'Resolved', enters manager verification credentials, and dispatches completion email alerts to the ticket reporter.
- Send Next Resolving Step (Orange Button): If the submitted proof is inadequate or if further mechanical/electrical adjustments are required, clicking this button returns the ticket back to the technician's queue accompanied by the manager's directives.

OPERATIONAL NOTE: Quality Assurance Protocol – Anti-Premature Closure Safeguard

To uphold campus infrastructure standards and prevent premature ticket closure, tickets cannot be closed directly by field technicians. Field workers submit proof-of-work documentation and photos ('Technician Solvation'), requiring mandatory review and sign-off by the Maintenance Administrator before the issue is certified as Resolved.

Technician Solution Submitted for Manager Review
 Technician Alex Johnson has submitted work notes and solution proof. Compare the Before & After results below and either confirm verification or send next step instructions.

#TKT-0007 REPORTED ON SEP 25, 2026, 04:57 PM

Main Library Library

LOCATION
 Prof. E. O. E. Pereira Mawatha, Peradeniya, Meewatura,
 Kandy District, Central Province, 20400, Sri Lanka
 Map Coordinates: 7.253°N, 80.593°E

REPORTED BY
 Minhaj Ali

REPORTED ISSUE DETAILS & PHOTO
 INITIAL PHOTO (REPORTED ISSUE)

REPORTED DESCRIPTION
 Broke

ASSIGN DUTY TECHNICIAN
 Select a qualified technician from the directory below to assign and dispatch this maintenance work order.

Alex Johnson (Facilities Management) — Active Shift

Assigned Technician: Alex Johnson Active Technician

BEFORE & AFTER RESOLUTION COMPARISON

BEFORE SOLVED (INITIAL REQUEST) Reported Issue

INITIAL PHOTO (BEFORE SOLVED)
 Broke

AFTER SOLVED (TECHNICIAN SOLUTION) Resolved Proof

RESOLVED PHOTO (AFTER SOLVED)
 Ok

MANAGER REVIEW & VERIFICATION OPTIONS
 Enter manager instructions or feedback below. Confirm & verify the task as completed, or send next step instructions back to the technician if further work is required.

Type manager feedback or next resolving step instructions for the technician here...

Confirm & Verify Resolved

Send Next Resolving Step

Figure 6.4: Maintenance Ticket Solvation Review — Geolocation & Initial Fault Overview (Part 1)

SECTION 7: FIELD TECHNICIAN / MAINTENANCE WORKER GUIDE

General Description of Functionality

The Field Technician Portal delivers a streamlined, mobile-responsive work order queue optimized for on-site maintenance technicians (alex). Technicians can view tasks assigned specifically to their trade, inspect manager directives, navigate to fault locations via GPS coordinates, and submit proof-of-work completion documentation.

7.1 Accessing the Technician Task Queue

General Description of Solution & Step-by-Step Procedure

Step 1 – Log in as Technician: Sign in using assigned technician credentials (alex / 1234).

Step 2 – Inspect Assigned Jobs: The TechnicianDashboard displays an uncluttered card deck of pending work orders classified by priority badge (Urgent, High, Medium, Low).

Step 3 – Commence Assignment: Click 'Inspect & Start Job' on the assigned ticket card.

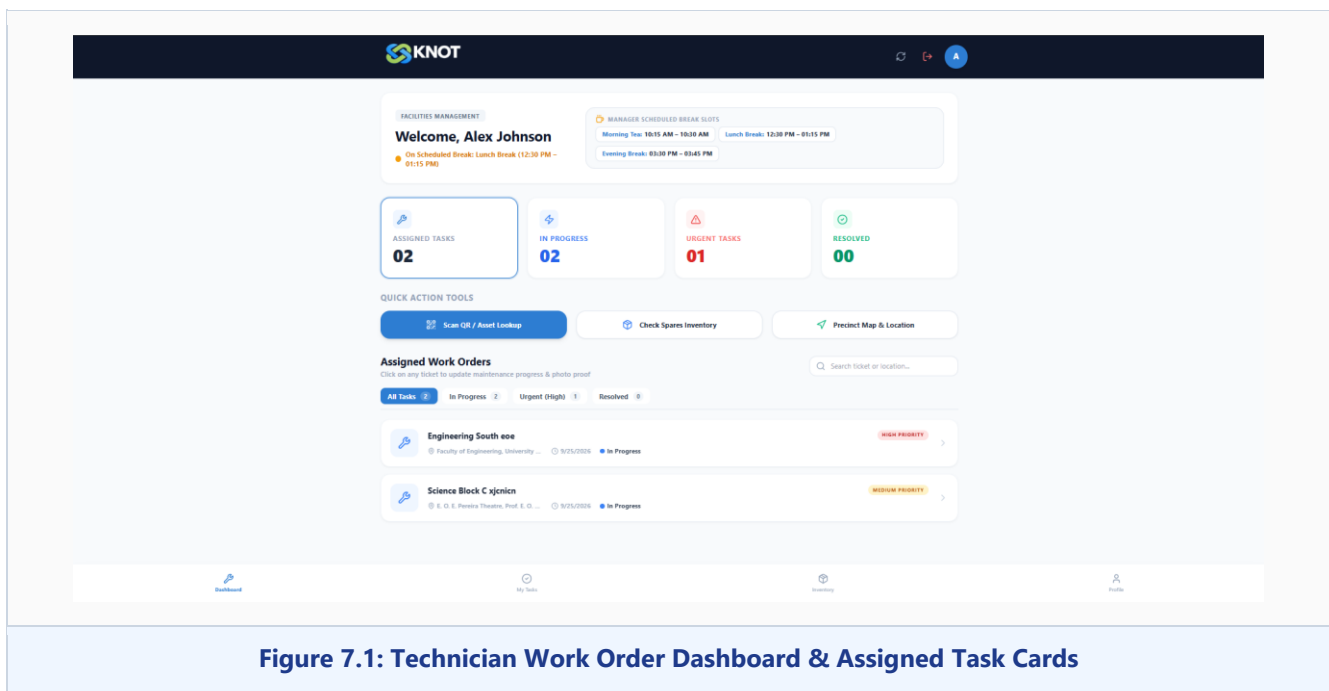


Figure 7.1: Technician Work Order Dashboard & Assigned Task Cards

7.2 Job Inspection, Work Log Completion & Proof-of-Work Logging

General Description of Solution & Step-by-Step Procedure

Step 1 – Review Work Directives: Examine the fault description, reporter contact info, location pin, and maintenance admin instructions.

Step 2 – Transition Job Status: Update ticket state from 'Open' to 'In Progress' upon arriving on site.

Step 3 – Execute Physical Repairs: Carry out required electrical, mechanical, or plumbing remediation.

Step 4 – Enter Resolution Work Notes: Input detailed technical remediation notes (e.g. 'Replaced faulty ballast capacitor and tested audio output across all 4 channels').

Step 5 – Upload Proof-of-Work Photo: Capture and upload a photograph demonstrating the completed, functional repair.

Step 6 – Submit Resolution: Set status to 'Resolved' and click 'Save Work Log'. The Works Admin and initial reporter receive automated closure notifications.

The screenshot displays a mobile application interface for 'Project KNOT'. A modal window titled 'Engineering South eoe' is open, showing a work log entry. The entry includes the following details:

- Location:** Faculty of Engineering, University of Peradeniya
- Reported by:** Minhaj Ali
- Issue Description:** broken
- Interactive Map:** A map showing the location with a blue pin. The GPS coordinates are 7.3544°, 80.3022°.
- Reported Issue Photo:** A placeholder image showing a blue cylindrical object with a white pipe and blue arrows indicating flow.
- Update Work Status:** Three buttons: 'OPEN', 'IN PROGRESS' (selected), and 'RESOLVED'.
- Maintenance Notes & Actions Taken:** A text input field containing the word 'done'.
- Proof of Completion Photo:** A section with a camera icon and the text 'Upload Completion Photo' and 'Click to capture or upload image proof'.
- Save Updates:** A blue button at the bottom of the modal.

Figure 7.2: Job Directive Inspection, Work Log Notes & Proof-of-Work Photo Upload

SECTION 8: AUTOMATED SYSTEM NOTIFICATIONS & ALERTS MATRIX

General Description of Functionality

Project KNOT operates an asynchronous notification subsystem that dispatches transactional email alerts and in-app status updates at critical operational milestones, ensuring strict transparency and regulatory compliance.

Event Trigger	Recipient Role	Notification Channel	Payload / Action Required
Student Request Submitted	Supervising Lecturer	In-App & Email	Notification to review and endorse or reject reservation.
Lecturer Endorsement Granted	Booking Admin (AR)	In-App & Email	Queued in AR Office portal for final institutional authorization.
Final Booking Approved	Student Booker	Email Confirmation	Official venue reservation receipt with booking reference code.
2-Step Verification Trigger	Student Booker	Banner & Email	Urgent prompt to confirm physical attendance before deadline cutoff.
Booking Auto-Cancelled	Student Booker & AR	Email Notification	Notice of cancellation and immediate release of hall slot due to no-show.
New Fault Ticket Logged	Works Engineering Admin	In-App & Email	Alert with category, description, and GPS coordinates for triage.
Work Order Dispatched	Assigned Technician	Mobile Web & Email	Task directive with priority level, location pin, and instructions.
Technician Solvation Submitted	Maintenance Admin	In-App & Email	Alert with Before/After proof to review and verify resolution or request next steps.
Task Verified & Resolved	Fault Reporter & Technician	In-App & Email	Formal sign-off confirmation and ticket closure notification.
Next Resolving Step Dispatched	Assigned Technician	Mobile Web & Email	Re-work directive with manager feedback and additional task instructions.

SECTION 9: FREQUENTLY ASKED QUESTIONS (FAQ) & TROUBLESHOOTING

Q1: What happens if I miss the 2-step verification deadline for my booking?

If attendance is not confirmed prior to the countdown expiration (12 hours before slot start for advance bookings, or 07:30 AM for same-day bookings), the automated anti-no-show engine permanently cancels the reservation, returns the room to public availability, and records an unverified strike against the booker's account.

Q2: Can undergraduate students reserve faculty venues without lecturer endorsement?

No. All student reservation requests require selecting an academic lecturer for supervisory endorsement. However, if the Booking Administrator has enabled the 'Auto-Booking Engine', conflict-free requests are automatically processed once endorsed.

Q3: Which academic venues are currently managed within Project KNOT?

The platform manages 10 standardized faculty venues: EOE Hall, Drawing Office 1 (DO1), Drawing Office 2 (DO2), Lecture Hall 01 (LH01), Lecture Hall 02 (LH02), Seminar Room A, Seminar Room B, Computer Lab 01, Computer Lab 02, and Electronics Lab.

Q4: What file specifications are required for bulk timetable importing?

The AR Office importer accepts standard Comma Separated Value (.csv) files. Headers must include: 'Room', 'Day', 'StartTime', 'EndTime', 'CourseCode', and 'Lecturer'. The validation parser automatically flags unrecognized venues or overlapping time blocks.

Q5: How does the reverse geocoding map feature work when reporting faults?

When you drop a pin on the interactive Leaflet map, the platform queries OpenStreetMap's Nominatim reverse geocoding API to resolve the latitude/longitude coordinates into a human-readable faculty building address, ensuring technicians arrive at the exact location.

SECTION 10: TECHNICAL SUPPORT & DIRECTORY

For access credential resets, special permission requests, or technical system assistance, please reach out to the designated faculty offices:

Faculty Office / Unit	Official Responsibility	Direct Contact Channel
Assistant Registrar (AR Office)	Master venue authorization, semester timetable scheduling & policy appeals	ar.office@eng.pdn.ac.lk Ext: 3004
Works Engineering Department	Campus physical infrastructure, electrical, plumbing & HVAC remediation	maint.admin@eng.pdn.ac.lk Ext: 3021
Faculty IT Center & KNOT Development	Platform hosting, single sign-on authentication, database & API issues	support@knotpdn.tech Room L03

— END OF PROJECT KNOT USER INFORMATION MANUAL (v2.5) —